# Yuncong Yang

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## Research Interests

- **World models and embodied AI** — object-centric representation, latent prediction under control conditioning, diffusion and flow-matching planners
- **Underwater perception** — visual SLAM, forward-looking imaging sonar, simulation-based evaluation
- **Underwater robot dynamics** — ROV–manipulator multibody modelling, grey-box residual correction of hydrodynamic models
- **Underactuated grasping** — linkage-driven gripper kinetostatics and dynamics, hold capacity, learning-based runtime prediction

## Education

**2024.09 – 2027.06 (expected)** M.Sc. in Civil and Hydraulic Engineering, Shanghai Jiao Tong University
School of Ocean and Civil Engineering; research on world models for underwater manipulation. Advisor: Dr. Xuyang Wang; cumulative GPA 3.22/4.0

**2020.09 – 2024.06** B.Eng. in Naval Architecture and Ocean Engineering, Dalian University of Technology
Comprehensive ranking 14/89 (top 15%); weighted average 86.7/100; GPA 3.67/4.0. Core coursework: Fluid Mechanics, CFD, Structural & Vibration Mechanics, Finite Element Analysis

## Research Experience

**World-model-based planning for deep-water salvage ROV ($C^3$-JEPA)**
*Master's thesis project · National Key R&D Program of China (2023YFC2809701) · 2024 – present*

- Developed the complete research pipeline, from data collection to closed-loop evaluation; benchmarked Dreamer-style latent imagination, DIAMOND diffusion models and pure JEPA before converging on $C^3$-JEPA, a cross-view, control-conditioned and context-extended design.
- Built a ROS–Gazebo data foundation (experimental pool, domain randomisation, teleoperation GUI) and collected a large synchronised multi-modal dataset across multi-camera vision, proprioception and all 12 thruster / 6 gripper control channels.
- Designed the observation module (frozen DINOv3-L features, temporal slot attention, SIGReg) and a lightweight object-centric predictor that fuses several camera views under control conditioning.
- Built a flow-matching diffusion planner with spherical-coordinate encoding and delay-compensated label shifting, plus a block-causal transformer rollout with critic-based trajectory selection.
- Learned slots isolate the task object and the grippers where unguided self-supervision fails outright; $C^3$-JEPA with the flow-matching planner cuts terminal positioning error by roughly a third and ranks candidate trajectories well above chance, with a complete salvage sequence running in closed loop.
- Trained every stage on a multi-GPU cluster with systematic ablations; results reported in the manuscript submitted to IEEE T-ASE (preprint arXiv:2609.30214).

**Underwater perception: visual SLAM and imaging sonar**
*Isaac Sim / Isaac Lab simulation study · 2026 · code*

- Rebuilt a pool-scale ROV scene in Isaac Sim 4.5 with four synchronised cameras, six AprilTag boards and a turbidity model (visibility, scattering, marine snow, caustics) driven by a live parameter dashboard, and recorded synchronised video with ground-truth poses.
- Built an offline visual SLAM chain — tag anchoring, KLT front-end, multi-view triangulation, PnP relocalisation and joint sparse bundle adjustment — reaching 97.1% / 93.3% localisable frames and 3.7 / 4.1 cm median error in clear and turbid water, measured in the tag world frame without ground-truth alignment or scale fitting.
- Implemented a forward-looking imaging sonar (ray casting, target strength, transmission loss, Rayleigh speckle, acoustic shadow) with pose-accumulated mapping; target range and position error were 0.25 m and 0.30 m inside the geometric window, complementing vision (0.09 m at 8 m) with 0.30 m ranging out to 12 m.

**Grey-box inverse dynamics for the salvage ROV**
*Branch study derived from the world-model project · 2024 – present*

- Formulated a physics-informed inverse-dynamics module coupling Fossen's 6-DOF equations with learned hydrodynamic parameters and a gated residual network, recovering thruster commands from observed vehicle motion and supplying the world model with an interpretable motion prior.

**Underactuated salvage gripper: kinetostatics, dynamics and runtime hold-capacity prediction**
*2024 – present*

- Built a virtual-work-based kinetostatic model of a double four-bar underactuated gripper giving closed-form contact-force and load-capacity solutions, and extended it to dry and immersed hold capacity, including sensitivity to the reading instant, object shape and seabed-assisted grasping.
- Validated the solver against ADAMS simulation and an experimental gripper prototype; derived the closed-form multibody dynamics via screw theory, resolving closed kinematic loops with non-minimal coordinates and Lagrange multipliers.
- Coupled the physical model with a parameter network and gated residual correction to absorb contact, friction and unmodelled hydrodynamic effects.
- Trained a residual contact-dynamics predictor on 20k simulated grasp episodes: 0.49–3.88 mm error over 20–150 ms horizons, against 1.44–63.3 mm for a constant-velocity baseline, across three seeds and held-out splits.
- Replaced a ten-step autoregressive rollout with a one-shot 100 ms predictor — 2.10 mm against 18.37 mm, 0.75 ms on CPU (8.6× faster) — reproducing the physical hold/slip criterion on 99.4–99.7% of starts.

## Publications

1. Y. Yang, et al. "Rapid Salvage of Deep-Sea Underactuated Manipulators: Real-Time Solution of Contact Forces and Impact Assessment of Disturbance Forces." *Robot* (in Chinese), accepted, 2025. EI-indexed.
2. Y. Yang, et al. "Underwater $C^3$-JEPA: An Object-Centric Cross-View World Model for ROV Salvage." Under review, *IEEE Transactions on Automation Science and Engineering* (T-ASE). Preprint: arXiv:2609.30214.
3. Y. Yang, et al. "Multibody Dynamics and Residual Correction for Underactuated Linkage Grippers." In preparation.
4. Y. Yang, et al. "Underwater Grasp Hold Capacity: From Design Boundaries to Non-Iterative Neural Prediction at Runtime." In preparation (target: *Ocean Engineering*).

## Patents

1. 1st inventor. Control System and Control Method for a Bionic Sea-Turtle Amphibious Vehicle. CN202310492163.5, filed 2023.
2. 4th inventor. Deep-Water Target Salvage Apparatus and Salvage Method. CN119329722B, granted 2025.
3. 4th inventor. Adaptive Underwater Adsorption Device and Underwater Salvage Apparatus. CN119262239B, granted 2025.
4. 4th inventor. Underwater Salvage Gripper Device and Salvage Apparatus. CN119260779A, filed 2024.
5. 8th inventor. Heavy Fuel Oil Heating System and Heating Method for Deep-Water Shipwrecks. CN121357744A, under examination.
6. 5th inventor. Bionic Sea-Turtle Flapping-Wing Locomotion Device. CN202310401637.0, filed 2023.

## Awards and Honors

- Second Prize, 12th National Ocean Vehicle Design and Manufacturing Competition (national final, C2 track)
- First Prize (E2, C2) and Third Prize (A1, E1), 12th Provincial Ocean Vehicle Competition; First and Third Prizes, 11th edition
- S Prize, Mathematical Contest in Modeling (MCM/ICM)
- MERIC Scholarship (2nd Class) and Academic Excellence Scholarship (2nd Class), Dalian University of Technology
- President, Lingdu Boat Team, Dalian University of Technology (2022 – 2024)

## Capabilities

- **First-principles modelling** — OpenFOAM CFD → Fossen 6-DOF identification → closed-form multibody dynamics for a closed-loop gripper (screw theory; validated in ADAMS and on a prototype).
- **End-to-end systems** — data capture (ROS–Gazebo, UUV Simulator, teleoperation GUI) → training on 8×A100 with systematic ablations → closed-loop evaluation; every component built from scratch.
- **Physics-based simulation** — an imaging sonar written from its acoustic model (ray casting, target strength, transmission loss, Rayleigh speckle, shadow); a turbidity-parameterised scene with marine snow and caustics as geometry.
- **Estimation** — visual SLAM (tag anchoring → joint sparse bundle adjustment); object-centric latent prediction and planning (JEPA, flow matching); errors stated without ground-truth alignment or scale fitting.
- **Hardware and tools** — STM32 embedded, motor drivers, serial links, gripper prototype assembly. Python / PyTorch, C, MATLAB, LaTeX; ROS / Gazebo, UUV Simulator, Isaac Sim 4.5; SolidWorks, ANSYS. Chinese (native); English (IELTS 6.5, October 2026).

## References

**Dr. Xuyang Wang**, Professor-level Senior Engineer — Underwater Engineering Institute, School of Ocean and Civil Engineering, Shanghai Jiao Tong University. wangxuyang@sjtu.edu.cn | +86 21 3420 8071

**Dr. Lei Qiao**, Tenure-track Associate Professor — Department of Naval Architecture and Ocean Engineering, School of Ocean and Civil Engineering, Shanghai Jiao Tong University. qiaolei@sjtu.edu.cn